

Minor-I Exam

Program: B. Tech. Branch: All Year: 1st Semester: 1st
Subject: Introduction to Computing (CS1001)

Date: 11/09/2025**Time Duration: 1.5 Hours****Start Time: 02:00 pm****Max. Marks: 40****Instructions:**

- 1) Attempt all questions carefully. All questions are mandatory.**
 - 2) Please don't copy from any other student. Plagiarism will be thoroughly checked for all copies.**
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Question 1: Represent the **sum** of these numbers in base 9: **(36)₇, (67)₈, (98)₁₀, and (34)₅**? [6]

Question 2: Convert decimal number **-18** to its 8-bit two's complement binary representation. [4]

Question 3: Represent **-30.675** in IEEE Single Precision format. [4]

Question 4: Explain the Basic C data types with suitable examples. [3]

Question 5: What is the difference between operators, division (/) and modulus (%), when used with integers and floating-point numbers, respectively? Justify your answer with suitable examples. [4]

Question 6: What is the value of the following expression? Note that the evaluation is on integers only. [4]

$$(!((4 - 4\%3) < 5 \ \&\& \ (6 / 4 > 3)))$$

Question 7: Which of the following statements does not have the same effect as all the others? Explain with reasons. [2]

(a) $k = k+1$; (b) $k = +1$; (c) $++k$; (d) $k+ = 1$;

Question 8: Write a one-line C expression to make the **least significant bit (LSB)** of an integer n as **0**. [3]

Question 9: Write the logical steps (in terms of expression(s), if needed) for **any one** of the following: [10]

- (a) Find the largest number between two given integers without using conditional and relational operators (use arithmetic / bitwise trick).
- (b) Given two positive integers a and b. Find their product after exchanging the last digits.
Input 4270 and 153, output 640950 ($4273 * 150$); Input 348 and 31, output 12958 ($341 * 38$).